

FineInstructions: Scaling Synthetic Instructions to Pre-Training Scale

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Abstract

- LLM's are trained on limited supervised data
 - This procedure is enhanced with smaller instruction-tuned data (examples of instruction and responses)
- Paper wants to overcome that problem by introducing the automatic generation of synthetic instruction and answer pairs
- Dataset: FineInstructions - ~18M instruction templates from real user data (queries and prompts)
- Instructions (Templates) is matched with human written documents
- This training procedure seems to outclass the traditional approach of self supervised word prediction learning
- [FineInstructions](#)

Introduction

Historical Derivation

- LLM's have traditionally been trained on “predict the next word” on large unstructured text data
 - Vast majority of time, computing and resources are spent on that task
- Supervised learning can enhance the capabilities with instruction - answer examples
 - This is called instruction tuning
 - Problem: Supervised instruction answer corpora are expensive and small
- Using Frontier Models (gpt etc.) used to generate instruction - answer examples
 - This approach only mimics the frontier model
- Conclusion:
 - In the end self supervised learning is the responsible for the models knowledge
 - Capabilities (answer a question, write a letter) are learned besides the factual knowledge
 - Question is: Is this the best way to train that kind of capabilities?
 - Alternative transformations pipelines (Manini et al. 2024; Su et al 2024) have been suggested

Approach in the paper

- Procedure called FineInstructions
 - Transformation of pre-training corpora into collections of instruction - response pairs
 - Generation of ~18M generic instruction templates derived from real user queries, which are subsequently instantiated on a per-document basis to create grounded Q&A pairs.
 - This procedure can be used for instruction tuning pre-model training
 - *Hypothesis*
 - This restructuring leads to better knowledge absorption and model performance
 - Simplify pre-training since low quality noise documents are handled before
 - Guarantees higher quality and more educational sections of a document
1. ~18M templates were created -> Represents highly diverse in-distribution synthetic data
 2. FineInstructions are introduced -> Converting pre-training documents into synthetic instruction - answer pairs
 3. Pre-training with FineInstructions outperforms standard pre-trainings

4. Release of a FineInstruction dataset of 1B+ synthetic instructions

[Figure 1. The FineInstructions pipeline for efficiently generating diverse, pre-training scale, synthetic instruction-answer pairs.](#)

Related work

- Genesis
 - Training on unstructured text
 - Apply filter method
 - Re-sampling, re-weighting
 - Converting raw data into instruction-answer pairs
 - Rephrasing low-quality documents
 - Conversion raw data into instruction -answer pairs
 - Approaches with instruction templates were usually hand crafted thus lack in raw numbers
- This paper
 - Generation of synthetic data by automatically creating and instantiating instruction templates

FineInstructions

Overview

- Collection of user-written queries -> questions, instructions, task requests
- Generalizing them into generic instruction templates
- Match between pre-training document with instruction templates
 - Match if answer to query is provided and sufficient for template instantiation
- A judge model is used to measure the quality
- Pipeline Logic:
 - Implements a “weak supervision” approach; transforms unlabeled corpora into supervised datasets via programmatic labeling.
 - Silver-Standard: Llama-3.3 70B (Teacher) generates high-quality examples.
 - Distillation: Efficient models (Llama-3.2 1B/3B) are trained on silver-standard data to replicate teacher performance.
 - Scaling: Distilled models generate 1B+ instruction-answer pairs at pre-training scale.

- Enable supervised pre-training from scratch.
- Ensures the model is “in-distribution” with downstream usage (prompt-response) from the first token.

Generating Instruction Templates

- Collection of human written prompt templates form various libraries
 - Filter of harmful queries
 - Decontamination (Filter “known” questions)
- Convert queries to templates by inserting tags for entities or szenarios
 - Within tags a description of what might be appropriate is inserted
 - Later the spans are filled from a matched document
- ~50k queries were selected for the fine tuning of a Llama-32 1B Instruction model
 - templates were converted to “silver” synthetic pipelines by “prompting an LLM with a series of prompts
 - Description (what kind of document could be instantiated and answer) have been created
- With tat ~18M templates were generated

Matching Documents to Instruction Templates

- Semantic model (BGE-M3) was used to embed all descriptions
- Corpus of unstructured documents where used to generate ~200k descriptions and embedded
- Store embeddings in FAISS (Fast semantic search db)
- Retrieve top 5 templates for a description
- Judge LLM labels for hard positives and hard negatives
- 2 step fine tuning of BGE-M3

Gaussian Pooling for Document Coverage

- Global mean pooling washes out specific information in long documents
- Solution: Generate K local embeddings (K=5) in addition to one global embedding
 - Centers are distributed linearly across the document length
 - Gaussian weights prioritize tokens near the center of each chunk
 - Training: BGE-M3 fine-tuned to reduce representation mixing and maintain local semantic clarity

- Validation: High Pearson correlation (0.99) between retrieved chunk index and actual answer location in the document

Retrieval

- Match candidates selected via cosine similarity threshold (> 0.865)
- Weighted random sampling used to encourage template diversity
- Templates vary in complexity: simple (1-2 tags) vs. complex (10+ tags)
- Sampling weights matched to complexity distribution of real-world queries (WildChat, LMSys, OAsst1, ShareLM)

Generating Instructions and Answers

Conclusion

[Figure 2. Efficiency from pre-training on FineInstructions data](#)

Models used in the paper

Model	Role / Designation	Primary Task	Purpose / Details
Llama-3.3 70B Instruct	Teacher Model	Silver-Standard Generation	Generates high-quality examples used for the distillation of smaller models.
Llama-3.2 1B Instruct	Query Genericizer	Template Creation	Transforms real-world user queries into generic instruction templates.
Llama-3.2 3B Instruct	Instantiator Model	Q&A Generation	Instantiates templates using source documents to produce grounded instruction-answer pairs.
BGE-M3	Embedding Model	Semantic Matching	Vectorizes documents and templates to

Model	Role / Designation	Primary Task	Purpose / Details
			enable efficient retrieval via FAISS.
Flow Judge (3.8B)	Final Judge	Quality Filtering	Evaluates generated pairs on a 5-point Likert scale; retains only high-quality samples (≥ 4).
Llama-3 Tokenizer	Tokenizer	Tokenization	Ensures consistent tokenization for the resulting pre-trained models.

Open Questions

- Entity and scenario replacement? How?
- Llama model pre-training “prompting an LLM with a series of prompts”
- Gaussian Pooling with only $K = 5$
- Pearson correlation a good choice? $k = 5$, semantic information centered in 1 and 5 (possibly?)